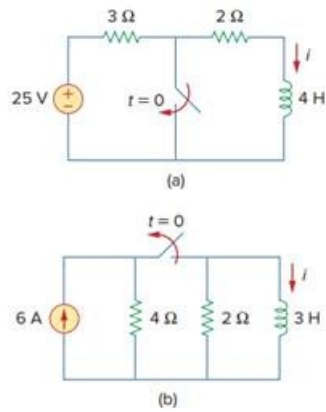


## Problem

Determine the inductor current  $i(t)$  for both  $t < 0$  and  $t > 0$  for each of the circuits in Fig. 7.119.

Fig. 7.119:

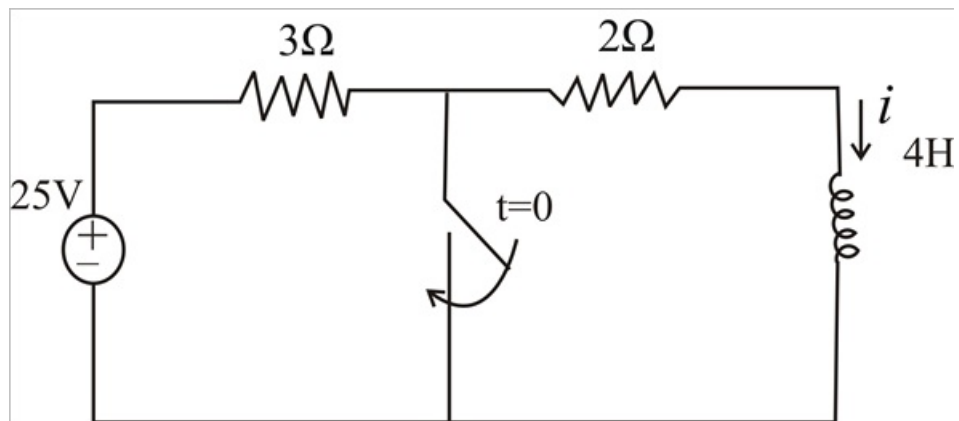


## Step-by-step solution

## Step 1 of 5

(A)

Given circuit diagram



## Step 2 of 5

For  $t < 0$  inductor act as short circuit and the switch is open

$$i = \frac{25}{3+2}$$

$$i = \frac{25}{5}$$

$$\boxed{i = 5\text{A}}$$

## Step 3 of 5

For  $t > 0$  the switch is closed, so time constant

$$\tau = \frac{L}{R}$$

$$\tau = \frac{4}{2}$$

$$\tau = 2$$

Therefore now the inductor current

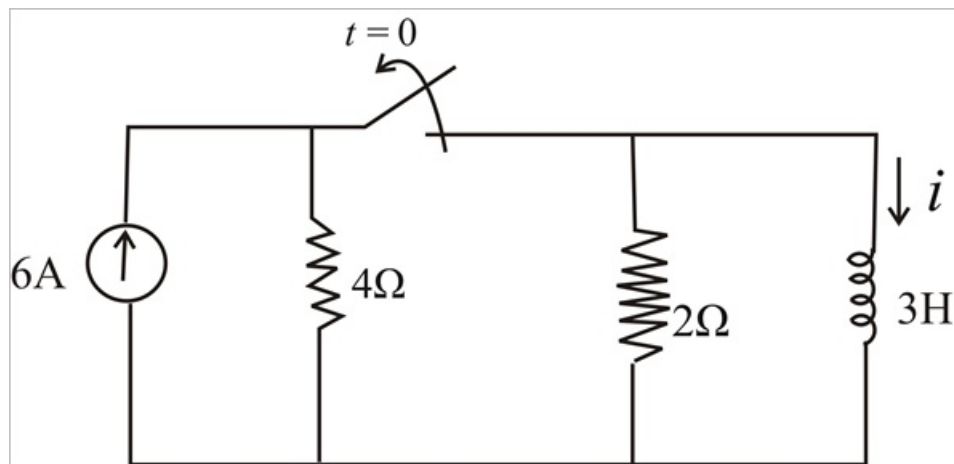
$$i(t) = 5e^{-\frac{t}{\tau}}$$

$$i(t) = 5e^{-\frac{t}{2}}u(t) \text{ A}$$

#### Step 4 of 5

(b)

Given circuit diagram



#### Step 5 of 5

For  $t < 0$  the switch is closed and inductor acts as short circuit

Therefore inductor current  $i = 6\text{A}$

For  $t > 0$  the switch is opened and time constant  $\tau = \frac{L}{R}$

$$\tau = \frac{3}{2}$$

Now the inductor current  $i(t) = 6e^{-\frac{t}{\tau}}$

$$i(t) = 6e^{-\frac{2t}{3}}$$

$$i(t) = 6e^{-\frac{2t}{3}}u(t) \text{ A}$$